

Analyzing Loan Default Risk: A Machine Learning Approach to Help Borrowers

How can machine learning models be used to predict and explain mortgage default risk, and how can data visualizations help borrowers and leaders understand the key factors driving that risk?



Aadit Razdan

Mentor: Dr. Pradip Nandi, Director - ECL Modeling and Analytics | BMO Bank

SPARK Research Internship 2025

Abstract

Loan defaults create major risks for both lenders and borrowers, making precise risk prediction vital for responsible lending. This project demonstrates how machine learning can guide more personalized lending choices and how feature-level insights can help both parties lower their financial risk.

Introduction

This project explores the use of supervised machine learning to predict the probability of mortgage default based on borrower and loan characteristics. This study utilizes Freddie Mac datasets. A comprehensive dataset is created by merging the Origination File — which contains characteristics such as credit score, DTI, CLTV, interest rate with the Performance File that tracks loan repayment outcomes over time. A machine learning model was developed to predict whether borrowers will default, estimate the probability of default, and identify key risk factors influencing these outcomes. In addition, exploratory data analysis and visualization were performed to reveal trends such as how default rates vary across economic drivers and to examine relationships between loan features and default risk. These combined approaches provide valuable insights that enhance risk assessment and support better decision-making for lenders and borrowers alike. Additionally, I developed a personalized risk assessment tool, providing targeted financial advice based on the individual's input profile. A visual risk meter displays risk levels for better interpretability.

Data Analysis

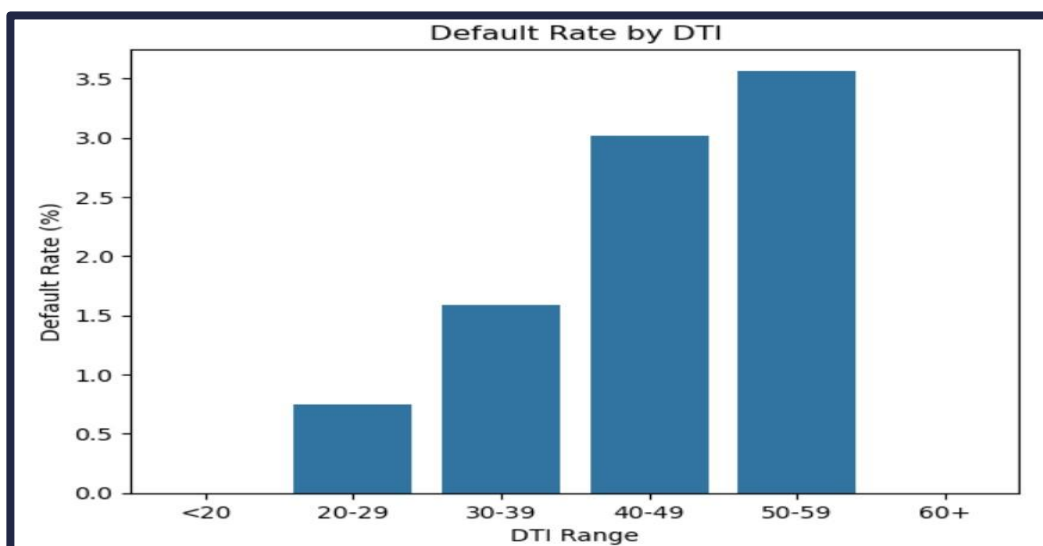


Figure 1: Default Rate directly proportional to DTI

The borrower has less disposable income available for living expenses and for absorbing unexpected financial shocks like illness or job loss. The borrower may face difficulties making timely debt payments.

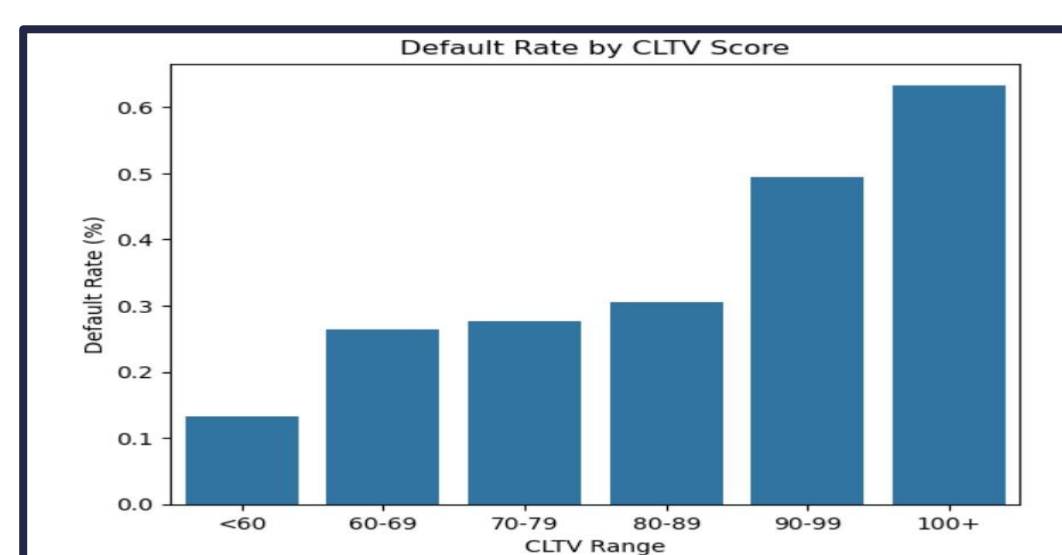


Figure 2: Default Rate directly proportional to CLTV Score

The borrower has less equity in the property, meaning they've financed a larger portion on the property's value. If this value declines, higher CLTV leads to loan balance exceeding property value.

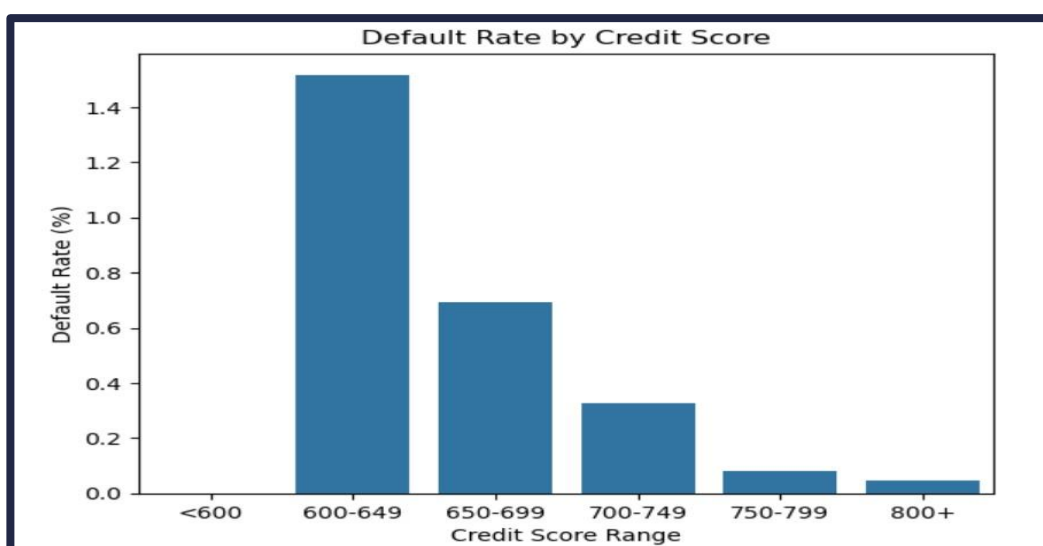


Figure 3: Default Rate inversely proportional to Credit Score

The higher the credit score is for the borrower, the more likely they are to be approved for loans. In addition, it allows borrowers to have more negotiating power to access more favorable terms and conditions, thus reducing default risk.

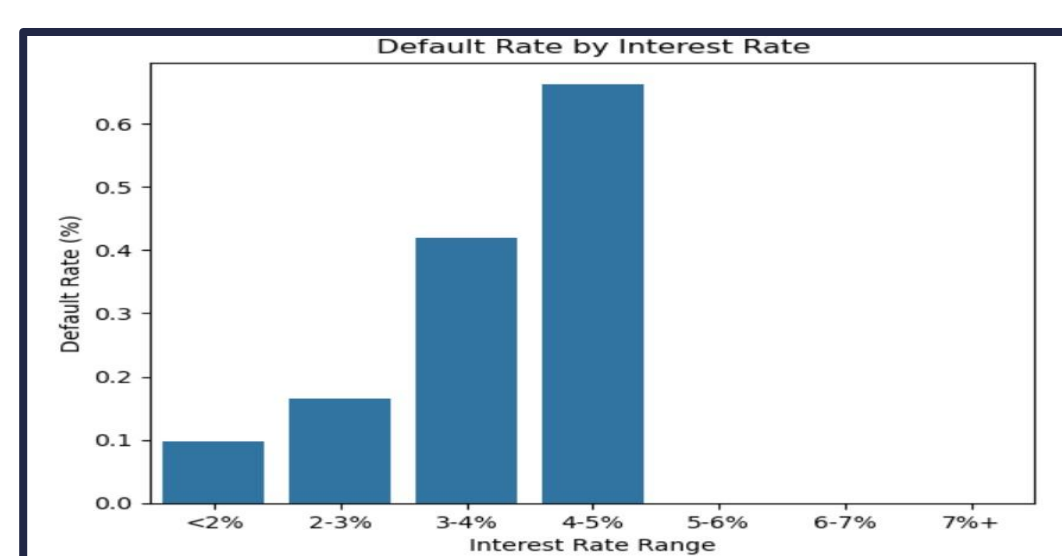


Figure 4: Default Rate is directly proportional to interest Rates

The higher that interest rates rise, the more that the dual challenges of increased financial hardship and higher borrowing costs negatively affect borrowers, thus increasing default risk.

Glossary

DTI - Debt to Income ratio - a financial metric used to assess an individual's ability to manage debt.

CLTV Score - represents the total revenue a business expects to generate from a single customer throughout their entire relationship with the company

Credit Score - A credit score is a numerical representation of your creditworthiness, based on the information in your credit report.

Interest Rate - The proportion of a loan that is charged as interest to the borrower, typically expressed as an annual percentage of the loan outstanding.

SHAP - SHapley Additive exPlanations is a game-theoretic approach used to explain the output of any machine learning model by assigning a "Shapley value" to each feature for a given prediction.

AUC - Area Under the Curve, is a crucial metric in machine learning, particularly for evaluating binary classification models.

Machine learning - Is a field of artificial intelligence that enables computers to learn from data without being explicitly programmed.

Supervised learning - Is a type of machine learning where an algorithm learns from a dataset that includes both input features and corresponding correct output labels.

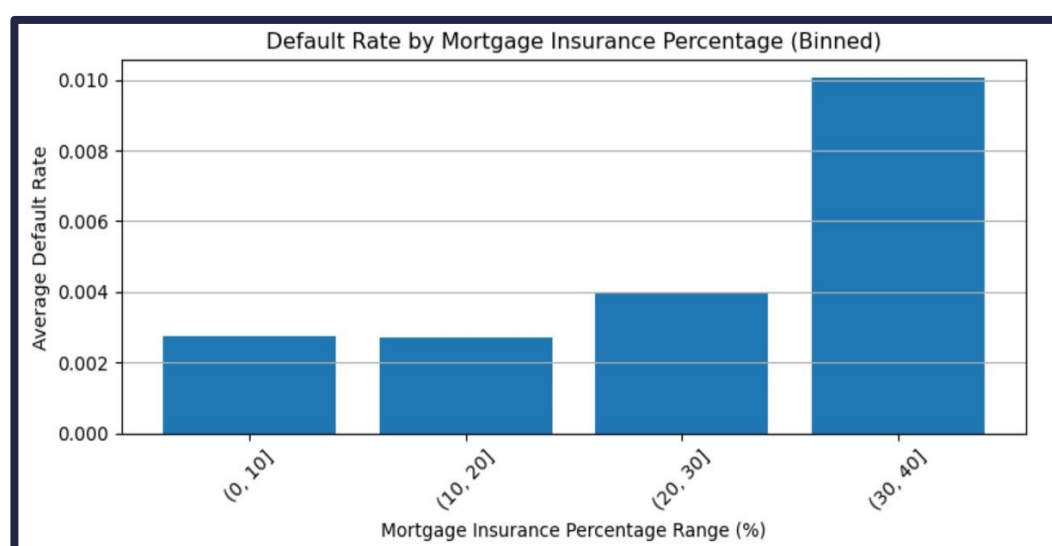


Figure 5: Default Rate directly proportional to Mortgage Insurance

While mortgage insurance enables borrowers to secure loans with smaller down payments, this trend suggests a challenge to borrowers' long-term financial stability and ability to maintain their mortgage obligations.

Methodology

The analysis began by acquiring Freddie Mac's 2021 Single-Family Loan-Level dataset. The Origination and Performance files were merged using unique loan identifiers to create a comprehensive dataset, and loans with insufficient performance history, less than 12 months, were filtered out to ensure data quality and accuracy.

```
# Inner join on a common column 'key'
merged_df = pd.merge(origination_df, df_filtered, on='Loan_Sequence_Number', how='inner')
num_rows = len(merged_df)
print("Rows in Merged Dataset ", num_rows)

# Sample data for graphs to run effectively
rows_to_sample = 1913039
sampled_df = merged_df.sample(n=rows_to_sample)
num_rows = len(sampled_df)
print("Rows in Sample Dataset ", num_rows)

Rows in Merged Dataset  1913039
Rows in Sample Dataset  1913039
```

Figure 6: Code snippet of merging datasets

Next, data cleaning and feature engineering were performed. Incomplete or inconsistent records were removed. Key features were derived, including loan-to-value ratio, payment-to-income ratio, number of missed payments, and credit score baskets. Categorical variables were encoded, and continuous variables were normalized to prepare the data for modeling.

```
# Feature engineering - Select features required for ML Model
features = ['Loan_Sequence_Number', 'Monthly_Reporting_Period', 'Current_Actual_UPB', 'credit_score', 'cltv_score',
            'dti', 'interest_rate', 'mortgage_insurance_percentage', 'num_borrowers', 'delinquency_flag']

final_df = sampled_df[features]

print("Final dataframe created")

Final dataframe created
```

Figure 7: Code snippet for feature engineering

Default was defined as loans with 90 or more days of delinquency. Each loan was labeled as default or non-default based on its repayment history to create the target variable for prediction on default.

```
df_filtered = performance_df[performance_df['Current_Loan_Delinquency_Status'] != 'RA']
df_filtered['Current_Loan_Delinquency_Status_int'] = df_filtered['Current_Loan_Delinquency_Status'].astype(int)

# Calculate Delinquency Flag
def calculate_delinquency(row):
    if (row['Current_Loan_Delinquency_Status_int'] >= 3):
        delinquency_flag = 1
    else:
        delinquency_flag = 0
    return delinquency_flag

df_filtered['delinquency_flag'] = df_filtered.apply(calculate_delinquency, axis = 1)
display(df_filtered.head(5))
```

Figure 8: Code snippet calculating delinquency flag

The dataset was then split into training and test sets using an 80/20 ratio. A Logistic Regression machine learning model was trained to predict the default status. Model performance was evaluated with metrics including accuracy, precision, recall, and the area under the receiver operating characteristic curve (AUC).

	precision	recall	f1-score	support
0	0.77	0.73	0.75	381932
1	0.74	0.79	0.76	381304
accuracy				763236
macro avg	0.76	0.76	0.76	763236
weighted avg	0.76	0.76	0.76	763236

ROC AUC: 0.8261730674067156

Figure 9: Code output showing model accuracy using precision, recall, F1-score and AUC

Finally, explainability techniques such as SHAP values were used to analyze feature importance. This enabled identification of the most influential risk factors driving default predictions and visualization of their impact on the model's outputs.

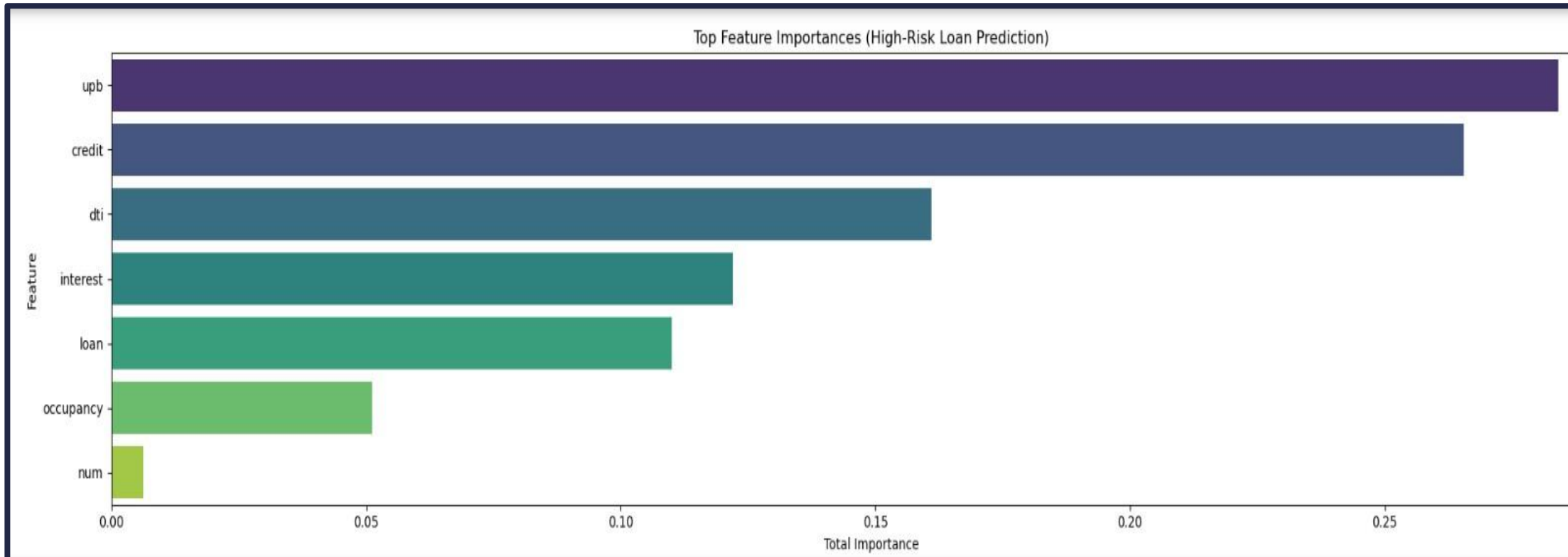


Figure 10: Code output showing feature importance using SHAP values

Results

Predicted default status: 1
Probability of default: 68.88%
High risk of loan default detected. Immediate financial review recommended.

Feature-based advice:

- * DTI of 50.00 is high. Consider any of the following steps to improve your DTI score:
- 1) Create a detailed monthly budget to track and control non-essential spending.
- 2) Pay down high-interest credit cards first to reduce monthly obligations.
- 3) Consider a side income stream, such as freelance work, part-time gigs, or selling unused items.

Borrower Risk Meter

Risk of Default: 0.0 0.2 0.4 0.6 0.8 1.0

Figure 11: Risk Meter - HIGH Default Risk

Predicted default status: 0
Probability of default: 46.33%
Moderate risk of default. You're not in danger, but some indicators raise concern.

Feature-based advice:

- * CLTV score of 88 suggests low equity. Consider any of the following steps to improve your CLTV score:
- 1) Making a larger down payment to lower the loan amount relative to the property value.
- 2) Paying down existing second liens or home equity loans to reduce the combined balance.
- 3) Waiting to borrow until your property value appreciates, improving the ratio in your favor.

Borrower Risk Meter

Risk of Default: 0.0 0.2 0.4 0.6 0.8 1.0

Figure 12: Risk Meter - MODERATE Default Risk

Predicted default status: 0
Probability of default: 3.88%
Low risk of default. Keep maintaining good financial habits.

Feature-based advice:

None - all financial indicators are within healthy ranges.

Borrower Risk Meter

Risk of Default: 0.0 0.2 0.4 0.6 0.8 1.0

Figure 13: Risk Meter - LOW Default Risk

Acknowledgements

I want to thank Dr. Pradip Nandi at BMO Bank for his consistent guidance throughout the duration of this project. I would also like to thank my SPARK peer reviewers, Saina Nayar and Alexander Gou, for their consistent feedback and helping me further my knowledge. And last, but certainly not least, I would like to thank Mr. Tobin and Mrs. Pasquesi for sponsoring the SPARK internship program at Adlai E Stevenson High School.

References

- Kling, A. (1970). Single Family Loan-Level Dataset. Freddie Mac. <https://www.freddiemac.com/research/datasets/sf-loanlevel-dataset>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (1970, January 1). *Scikit-Learn: Machine learning in Python*. Journal of Machine Learning Research. <https://www.jmlr.org/papers/v12/pedregosa11a.html>
- Hunter, J.D. (2007) Matplotlib: A 2D Graphics Environment. Computing in Science & Engineering, 9, 90-95. <http://dx.doi.org/10.1109/MCSE.2007.55>
- McKinney, W. (2010, May 1). Data Structures for Statistical Computing in python - scipy proceedings. scipy. <https://proceedings.scipy.org/articles/Majora-92bf1922-00a>

Conclusion

My machine learning model predicts mortgage default risk with high accuracy by combining origination and performance data. Key features such as credit score, debt-to-income ratio (DTI), combined loan-to-value (CLTV), interest rate, mortgage insurance percentage, and the number of borrowers proved to be strong indicators of default. Merging the datasets strengthened the predictive power of the model, while trend visualizations revealed clear patterns connecting borrower profiles to default outcomes.

Predicted default status: 0
Probability of default: 49.85%
Moderate risk of default. You're not in danger, but some indicators raise concern.

Feature-based advice:

None - all financial indicators are within healthy ranges.

Borrower Risk Meter

Risk of Default: 0.0 0.2 0.4 0.6 0.8 1.0

Control Case:

Credit Score: 700, CLTV: 78,
DTI: 38, Int Rate: 2.9%, MI: 20%
of Borrowers: 2
Default Probability: 49.85%
Figure 14: Risk Meter - Control Case

Experiment 1:
Credit Score increases to 710
Default Risk decreases
Default Probability: 44.43%

Figure 15: Risk Meter - Experiment 1

Predicted default status: 1
Probability of default: 57.16%
Moderate risk of default. You're not in danger, but some indicators raise concern.

Feature-based advice:

- * CLTV score of 88 suggests low equity. Consider any of the following steps to improve your CLTV score:
- 1) Making a larger down payment to lower the loan amount relative to the property value.
- 2) Paying down existing second liens or home equity loans to reduce the combined balance.
- 3) Waiting to borrow until your property value appreciates, improving the ratio in your favor.

Borrower Risk Meter

Risk of Default: 0.0 0.2 0.4 0.6 0.8 1.0

Experiment 3:
DTI increased to 48
Default Risk increases
Default Probability: 57.47%

Figure 17: Risk Meter - Experiment 3

Predicted default status: 1
Probability of default: 64.46%
High risk of loan default detected. Immediate financial review recommended.

Feature-based advice:

- * Interest rate of 3.00% is above average. Consider any of the following steps to lower your interest rate:
- 1) Refinancing your loan with another lender offering better rates - especially if your credit score has improved.
- 2) Shopping around and comparing at least 3+ lenders to find the lowest available interest rates.
- 3) Increasing your down payment or reducing loan amount if possible, as this can qualify you for better rates.

Borrower Risk Meter

Risk of Default: 0.0 0.2 0.4 0.6 0.8 1.0

Experiment 5:
Mort. Ins. increased to 30%
Default Risk decreases
Default Probability: 49.68%

Figure 19: Risk Meter - Experiment 5

Predicted default status: 0
Probability of default: 31.77%
Low risk of default. Keep maintaining good financial habits.

Feature-based advice:

- * Number of borrowers is 3.0, which may introduce more complexity to the loan.
- Consider any of the following steps to strengthen your loan profile:
- 1) Ensure all borrowers have solid credit scores and stable income - lenders may focus on the weakest profile.
- 2) If any borrower has poor credit or low income, explore whether they can be removed from the application.
- 3) Clearly define financial responsibilities among borrowers to prevent confusion or missed payments.

Borrower Risk Meter

Risk of Default: 0.0 0.2 0.4 0.6 0.8 1.0

Experiment 6:

of Borrowers increased to 3
Default Risk decreases
Default Probability: 25.77%

Figure 20: Risk Meter - Experiment 6